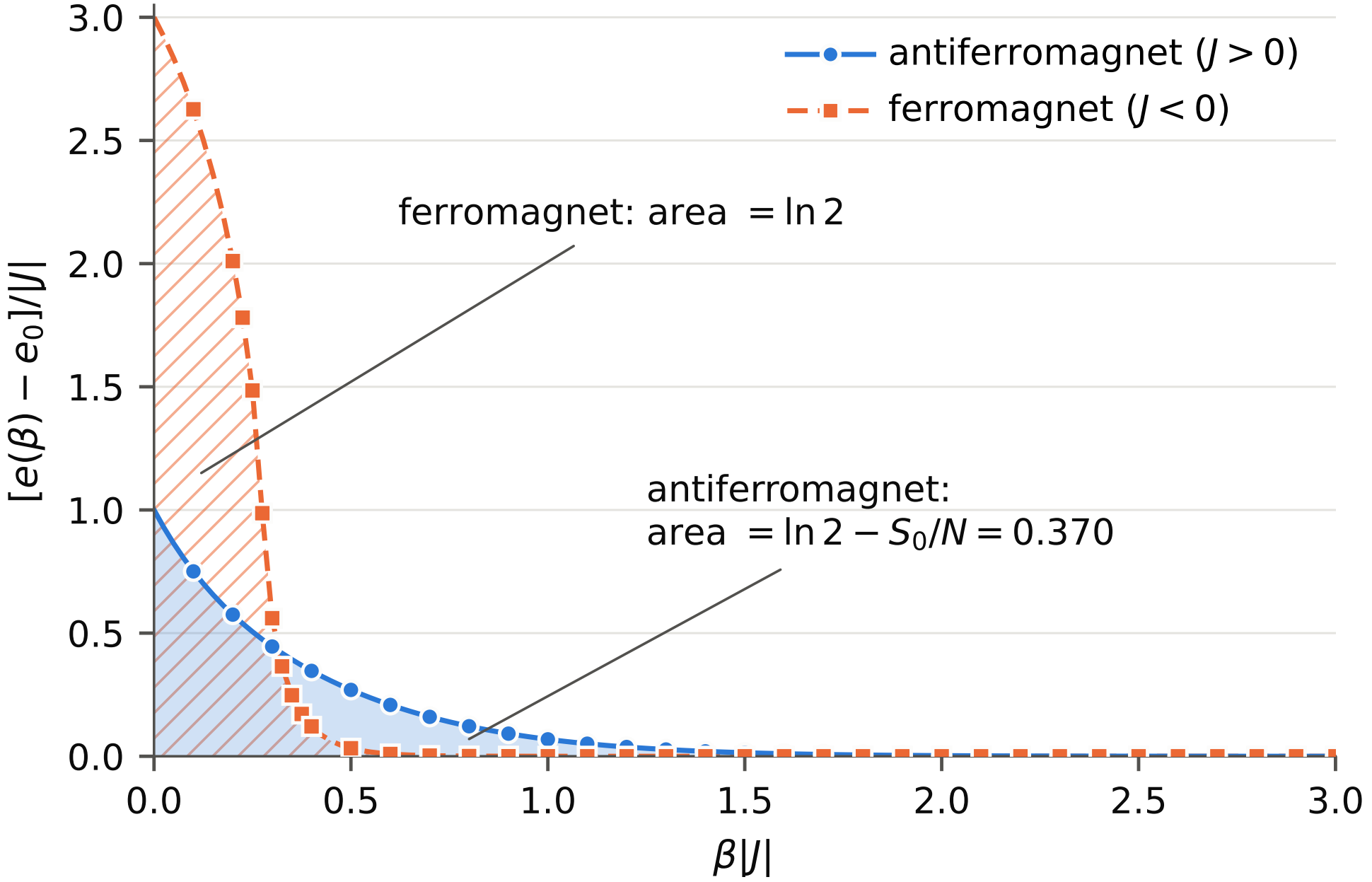
antiferromagnet ($J > 0$)
ferromagnet ($J < 0$)
ferromagnet: area $= \ln 2$
antiferromagnet:
area $= \ln 2 - S_0/N = 0.370$
$[e(\beta) - e_0]/|J|$
$\beta|J|$
0.0
0.5
1.0
1.5
2.0
2.5
3.0